NES AI FINANCE & INSURANCE INTELLIGENCE PLATFORM

Data Model & ERD Specification

Document ID: NES-DATA-008 | **Version:** 1.0

Build approach: Cline-assisted development • Local-first Windows application

Scope: Personal research, paper trading and insurance document assistance

Proposed entities, relationships and integrity rules

1. Modelling principles

- Separate source data, calculated results and user-confirmed records.
- Record timestamps and provenance for market data and extracted policy fields.
- Use stable identifiers; avoid using display names as primary keys.
- Protect sensitive fields and avoid storing unnecessary personal information.
- Version backtest configurations and model/strategy parameters.

2. Proposed entities

Entity	Key fields / relationships
UserPreferences	theme, reduced_motion, default_exchange, timezone, layout preferences.
DataSourceConfig	provider_name, enabled, status, last_success_at, settings_reference; never store plain secrets in UI data.
Watchlist	id, name, created_at; one-to-many WatchlistItem.
WatchlistItem	id, watchlist_id, symbol, exchange, order_index, added_at.
MarketBar / QuoteCache	symbol, exchange, timestamp, OHLCV or quote fields, source, quality/status.

Entity	Key fields / relationships
Strategy	id, name, version, rules_json, parameters_json, created_at.
BacktestRun	id, strategy_version, universe, period, assumptions, status, metrics_json, created_at.
PaperPortfolio	id, name, initial_cash, current_cash, currency, created_at.
PaperOrder	id, portfolio_id, symbol, side, quantity, simulated_price, status, timestamp; simulation only.
PaperPosition	portfolio_id, symbol, quantity, average_price, realised/unrealised values where supported.
Policy	id, category, provider, policy_reference, dates, premium, coverage fields, verification_status.
PolicyDocument	id, policy_id, local_path_reference, checksum, mime_type, processing_status, created_at.
ExtractedField	document_id, field_name, raw_value, normalized_value, provenance, confidence_if_supported, review_status.
PolicyComparison	id, policy_ids, comparison_fields, missing_fields, generated_at.
Reminder	id, related_record_id, reminder_date, status, notification_preferences.
AIResearchItem	id, query, answer, source_refs, limitations, model_id, created_at.
AuditEvent	id, event_type, entity_type, entity_id, timestamp, safe_metadata.

3. Relationships

- Watchlist has many WatchlistItems.
- Strategy has many versioned BacktestRuns.
- PaperPortfolio has PaperOrders and PaperPositions.
- Policy may have multiple PolicyDocuments.
- PolicyDocument has many ExtractedFields, each with review status and provenance.
- PolicyComparison references two or more Policies.
- Reminder links to a policy or other supported record.

4. Data integrity

- Use decimal-safe representation for money where appropriate; avoid binary floating point for persisted currency calculations.
- Use timezone-aware timestamps internally and display local time consistently.
- Validate quantity, date ranges, currency and category fields.
- Do not treat unverified OCR values as authoritative.
- Document migration strategy and test rollback/restore where supported.

5. ERD and implementation note

Create an ERD before implementing the real database. The entities above are a proposed logical model, not a final schema. Confirm provider requirements, document retention and actual workflows before migration scripts are written.